

Irene Balelli

Ph.D.

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Biographical Information

Birth **November 3, 1988**, in Forlimpopoli - Italy.
Citizenship **Italian and French**.
Family sit. **Married. Two children** (2015 and 2018).

Research Interests

Modeling/ Statistics Bayesian Statistics. Latent variable models. Generative models. Mixed effects models. Identifiability analysis. Sensitivity analysis. PK/PD. In-silico trials. Meta-modeling.

ML & Data governance Causal learning. Graph-based learning. Federated learning. Distributed learning. Differential privacy. Large-scale heterogeneous biomedical data. Missing data.

Biomedical Applications *Cardiology*: Drugs' cardiac safety. Pro-arrhythmic risk. Cardioembolic stroke. ECG data.
Neurodegenerative diseases: Alzheimer's Disease. Parkinson's Disease. Brain Imaging data.
Genetic-related diseases: FragileX Syndrome. Cancer. Multi-omics data.
Immunology: Immune system. Antibody affinity maturation. B-cells. Immune memory. Vaccination.

Probability Random walks on graphs. Markov chains. Graph theory. Galton-Watson processes.

Current Position & Work experience

2021–now **Research scientist (ISFP) in mathematical modeling for computational biomedicine**, EPIONE team - Inria Center at University Côte d'Azur, Valbonne - France.

2020–2021 **Post-doctoral research fellowship in federated statistical learning for biomedical data**, EPIONE team - Inria Center at University Côte d'Azur, Valbonne - France.

2017–2019 **Post-doctoral research fellowship in modeling of the immune response to Ebola vaccine**, SISTM team - Inserm U1219 Bordeaux Population Health, Bordeaux - France.

Education

2017 **Qualification**, CNU section 26: Applied mathematics and applications of mathematics.

2013–2016 **PhD in Applied Mathematics**, LAGA - University Paris 13, France, PhD thesis: *Mathematical foundations of antibody affinity maturation* (Supervisors: V. Milišić, G. Wainrib, H. Zaag).

2011–2013 **Master Degree in Mathematics Applied to Biology and Medicine**, University Paris 6, France, Master thesis: *A mathematical model of somatic hypermutation* (Supervisors: V. Milišić, G. Wainrib).

2010–2011 **Erasmus Program**, Universidad Complutense, Department of Mathematics, Spain.

2007–2010 **Bachelor degree in Mathematics**, University of Bologna, Italy, Bachelor thesis: *Bressloff and Cowan's model: visual hallucinations as stable states of cortical activation* (Supervisor: G. Citti).

Scientific production →



International journals

- W. Khatir, M. Lorenzi, **I. Balelli**, C. Gwizdek, *MCVAE-based multi-omic anomaly detection in Fragile X Syndrome*, Accepted in Nucleic Acid Research - Molecular Medicine 2026. [[hal-05639655](https://hal.archives-ouvertes.fr/hal-05639655)].

- S. Al-Ali, **I. Balelli**, *Multi-Channel Causal Variational Autoencoder for multimodal biomedical causal disentanglement*, Journal of Biomedical Informatics, Vol. 176, 104995, 2026. [[hal-05483061](#)].
- V. Carbone, **I. Balelli**, Y. Coudière, O. Camara, B. Trénor, M. Barbier, M. Sermesant, *A cloud-based platform for seamless digital testing of cardiac medical devices and drugs*, Project Repository journal (PRj), Vol. 24, 2025, [DOI](#) [[PDF](#)].
- Q. Clairon, C. Pasin, **I. Balelli**, R. Thiébaut, M. Prague, *Parameter estimation in nonlinear mixed effect models based on ordinary differential equations: an optimal control approach*, Computational Statistics, 2024, 39(6), 2975-3005 ([COST](#)).
- [†]**I. Balelli**, S. Silva, M. Lorenzi, *A Differentially Private Probabilistic Framework for Federated Heterogeneous Multi-View Datasets Variability*, Journal of Machine Learning for Biomedical Imaging ([MELBA](#)). 2022:012.
- **I. Balelli**, C. Pasin, M. Prague, F. Crauste, T. Van Effelterre, V. Bockstal, L. Solforosi, R. Thiébaut, *A model for establishment, maintenance and reactivation of the immune response after vaccination against Ebola virus*, [Journal of Theoretical Biology](#), 2020, DOI: 10.1016/j.jtbi.2020.110254.
- C. Pasin, **I. Balelli**, T. Van Effelterre, V. Bockstal, L. Solforosi, M. Prague, M. Dougouih, R. Thiébaut, *Dynamics of the humoral immune response to a prime-boost Ebola vaccine: quantification and sources of variation*, [Journal of Virology](#), 2019, DOI: 10.1128/JVI.00579-19.
- **I. Balelli**, V. Milišić, G. Wainrib, *Multi-type Galton-Watson processes with affinity-dependent selection applied to antibody affinity maturation*, [Bulletin of Mathematical Biology](#), 2019, vol. 81, no 3, p. 830-868.
- **I. Balelli**, V. Milišić, G. Wainrib, *Random walks on binary strings applied to the somatic hypermutation of B-cells*, [Mathematical Biosciences](#), 2018, vol. 300, p. 168-186.

Scientific book chapters

- **I. Balelli**, S. Al-Ali, E. Dumas, J. Abecassis, *Causality: fundamental principles and tools*, Trustworthy AI in Medical Imaging, 2024, MICCAI/Elsevier book series, [[hal-04831368](#)].
- [†]F. Cremonesi, M. Vesin, S. Cansiz, Y. Bouillard, **I. Balelli**, L. Innocenti, S. Silva, S.S. Ayed, R. Taiello, L. Kameni, R. Vidal, F. Orhac, C. Nioche, N. Lapel, B. Houis, R. Modzelewski, O. Humbert, M. Onen, and M. Lorenzi, *Fed-BioMed: Open, Transparent and Trusted Federated Learning for Real-world Healthcare Applications*, To appear in: Federated Learning Systems - Towards Privacy-Preserving Distributed AI, Springer, 2025, [[hal-04081557](#)].
- M. Lorenzi, M. Deprez, **I. Balelli**, A. L. Aguila, A. Altmann, *Integration of Multimodal Data*, Machine Learning for Brain Disorders, 2023, 573-597, Springer Nature.

International peer-reviewed conferences with proceedings

- A. Lang, R. Ramos, S. Al-Ali, M. Mousavi, A. Calissano, **I. Balelli**, *Consensus Enhances Individual Causal Models: a Use Case on Lung Cancer Driven by Cellular Pathways*, 24th International Conference on Computational Methods in Systems Biology (CMSB) 2026. [[hal-05620648](#)].
- [†]S. Al-Ali, J. Rodríguez Padilla, M. Sermesant, **I. Balelli**, *Cardiac Electromechanical Model Sensitivity Analysis using Causal Discovery*, Functional Imaging and Modeling of the Heart (FIMH) 2025. Lecture Notes in Computer Science, 15672. Springer, Cham. DOI [[hal-05121540](#)].
- [†]S. Al-Ali, M. T. Mora, M. Sermesant, B. Trénor, **I. Balelli**, *Assessing ion channel blockade and electromechanical biomarkers' interrelations through a novel Multi-Channel Causal Variational Autoencoder*, Computing in Cardiology (CinC) 2024. 51, 1 ISSN: 2325-887X DOI, [[hal-04607082](#)].
- Y. Coudière, M. Leguèbe, **I. Balelli**, A. Baretta, G. Fauré, D. Feuerstein, *A computer model for in-silico trials on pacemaker energy efficiency*, Computing in Cardiology (CinC) 2024, [[hal-04886140](#)].
- [†]S. Al-Ali, J. Llopis-Lorente, M. T. Mora, M. Sermesant, B. Trénor, **I. Balelli**, *A causal discovery approach for streamline ion channels selection to improve drug-induced TdP risk assessment*, IEEE Xplore2023 - Computing in Cardiology (CinC) 2023 [[hal-04105144](#)].
- [†]**I. Balelli**, S. Silva, M. Lorenzi, *A Probabilistic Framework for Modeling the Variability Across Federated Datasets of Heterogeneous Multi-View Observations*, International Conference on Information Processing in Medical Imaging ([IPMI](#)). Springer, Cham, 2021. p. 701-714.

National and international conferences without proceedings

- A. Baretta, Y. Coudière, O. Camara, B. Trenor, H. Arevalo, **I. Balelli**, R. Setzu, L. Geris, S. Benito, M. Barbier, M. Sermesant, *SimCardioTest cloud-based platform: an innovative tool for cardiac in silico trials*, Abstract, VPHi Conference 2024.
- Mélanie Prague, Jeffrey M Gerold, **I. Balelli**, Chloé Pasin, Jonathan Z Li, Dan H Barouch, James B Whitney, and Alison L Hill, *Viral rebound kinetics following single and combination immunotherapy for HIV/SIV*, Conference on Retroviruses and Opportunistic Infections (CROI). Poster. Boston, USA, 2018.

Others

- [†]**I. Balelli***, A. Sportisse*, F. Cremonesi, P.A. Mattei, M. Lorenzi, *Fed-MIWAE: Federated Imputation of Incomplete Data via Deep Generative Models*, In preparation, [[hal-04069795](https://hal.archives-ouvertes.fr/hal-04069795)].
- M. Prague*, J. Gerold*, **I. Balelli**, C. Pasin, J. Li, D. Barouch, J. Whitney, A. Hill, *Viral rebound kinetics following single and combination immunotherapy for HIV/SIV*, [[bioRxiv 700401](https://doi.org/10.1101/2023.07.10.536401)].
- **I. Balelli**, V. Milišić, G. Wainrib, *Branching random walks on binary strings for evolutionary processes in adaptive immunity*, [[arXiv: 1607.00927](https://arxiv.org/abs/1607.00927)].

[†] This paper is listed within the 3IA Côte d'Azur collection.

Conferences, Meetings and Seminars

- 2026 **Mathematical Foundations of AI - When Uncertainty Quantification meets Causal Modelling and Inference**, *Plenary talk*, SCAI - Paris.
From Cell to Clinic: Multi-scale Mechanistic Modeling and Statistical Learning for Oncology, *Invited talk*, Lyon.
MoMI - Le Monde des Mathématiques Industrielles, *Invited talk*, Sophia Antipolis.
- 2024 **IABM 2024**, *Invited talk*, Grenoble.
i2m Seminar, *Invited talk*, Marseille.
HeKa Seminar, *Invited talk*, Online.
- 2023 **Innovaheart 2023**, *Invited talk*, Bordeaux.
3IA Seminar, *Invited talk*, Sophia Antipolis.
- 2022 **Inria-DFKI workshop 2022**, Bordeaux.
BoostUrCareer Doctoriales 2022 – AI in HEALTHCARE, *Invited talk*, Nice.
- 2021 **Information Processing in Medical Imaging (IPMI) 2021**, Online event.
- 2020 **3IA Scientific Days**, Nice - France.
Sophi.A Summit 2020, Sophia Antipolis - France.
- 2019 **4th EBOVAC1/2 Annual meeting**, *Invited talk*, Nairobi - Kenya.
VRI Annual meeting, *Invited talk*, Paris - France.
- 2018 **IMI 10th Anniversary Scientific Symposium**, [3rd committee prize “best presentation”](#), Brussels - Belgium.
CROI 2018, *Poster (J.M. Gerold, C. Pasin, I. Balelli, S. Lim, C. Osuna, J.B. Whitney, D.H. Barouch, M. Prague, A.L. Hill)*, Boston - United States.
3rd EBOVAC1/2 Annual meeting, *Invited talk (with C. Pasin)*, Amsterdam - Nederland.
- 2017 **Systems Immunology and Vaccine design**, Heidelberg - Germany.
- 2016 **Probabilities and Statistics seminar (LAGA)**, *Invited talk*, Villetaneuse - France.
1st Challenges in inflammation meeting, Florence - Italy.
Les probabilités de demain, *Invited talk*, IHÉS - Bures-sur-Yvette - France.
Summer school: “PDE and Probability for Life Sciences”, CIRM - Marseille - France.
- 2015 **EDP-Normandie**, Havre - France.

- 2014 **InflaConf: Mathematical modeling in immunology and inflammation**, *invited talk*, Paris - France.
CANUM 2014, Carry-le-Rouet - France.
Inflamex day, *Invited talk*, CIEP Sèvres - France.
- 2013 **GDR Métice: Inflammation and Treatment Resistance**, *invited talk*, Lyon - France.

Thematic schools: contribution

- 2022 **AI4Health Winter School**, *Workshop (teaching materials and presentation)*: Fed-BioMed, an open source framework for federated learning in real world healthcare applications, Online event.
- 2021 **AI4Health Winter School**, *Workshop (teaching materials and presentation)*: Handling heterogeneity in the analysis of biomedical information, Online event.
- First Inria-DFKI European Summer School on Artificial Intelligence**, *Workshop (teaching materials and presentation)*: Federated learning methods and frameworks for collaborative data analysis, Online event.

Supervision activities

- Post-Doc **S. Al-Ali**, *Causal data analysis of in-silico trials*, Oct. 2022 - July 2025 (100% supervision).
- PhD **W. Kahtir**, *Integromics analysis: a new angle for studying the pathophysiology of Fragile X Syndrome*, co-directed with C. Gwizdek (IPMC) and M. Lorenzi, Since Mar. 2024 (33% supervision)[‡].
- A. Bakhouché**, *Analysis of European National Health data and vascular registries to better understand the outcomes of patients with vascular diseases*, co-directed with H. Delingette and J. Raffort-Lareyre (CHU Nice), Since Sept. 2025 (33% supervision)[‡].
- A. Lang**, *Multimodal prediction for cardioembolic stroke: etiology and risk*, co-directed with M. Lorenzi, Since Dec. 2025 (50% supervision)[‡].
- E. Gaymard**, *Innovative mathematical methodologies in pharmacometric meta-modeling from highly heterogeneous sources*, industrial PhD (CIFRE) with [Exact-Cure](#), co-directed with M. Sermesant and S. Benito (ExactCure), Nov. 2022 - Jun. 2023 (33% supervision).
- Master 2 **A. Lang**, *Voting strategy for causal discovery with uncertainty*, Apr. 2025 - Sept. 2025 (100% supervision).
- B. Ramudu Manam**, *PPCA-based disease progression modeling*, Apr. 2023 - Sept. 2023 (50% supervision).
- A. Senacheribbe**, *Longitudinal PPCA*, Sept. 2020 - Feb. 2021 (50% supervision).
- M. Alexandre**, *Mechanistic modeling for vaccinology*, Feb. 2019 - Aug. 2019 (20% supervision).
- Master 2, **I. Illahibuccus-Sona**, *Semi-supervised multi-modal VAE*, Nov. 2024 - Apr. 2025 (100% supervision).
- Res. Proj. **H. Hammed**, *Causal inference with application to drug-induced proarrhythmic risk*, Nov. 2024 - Apr. 2025 (50% supervision).

[‡] I am committed with the doctoral school [ED SVS](#) and UniCA to defend my HDR by 2026.

Teaching Activities

- 2025-2026 **Causal Learning for Healthcare**, *Lectures and tutorials (18h/year)*, M2 MSc Data Science & Artificial Intelligence - Health track, University Côte d'Azur, Sophia Antipolis - France.
- 2023-2025 **Advanced statistical modeling**, *Lectures and tutorials (20h/year)*, 3rd year bachelor's degree (BUT Data science), University Côte d'Azur, Sophia Antipolis - France.
- Statistical modeling for complex data and Big Data**, *Lectures and tutorials (33h/year)*, 3rd year bachelor's degree (BUT Data science), University Côte d'Azur, Sophia Antipolis - France.
- Research awareness**, *Talks and demos (6h/year)*, M1-M2, EURECOM, Sophia Antipolis - France.
- 2020-2024 **Bayesian learning**, *Lectures and tutorials (10h/year)*, M2 MSc Data Science & Artificial Intelligence, University Côte d'Azur, Sophia Antipolis - France.

- 2020-2023 **Modeling of biological systems**, *Lectures and tutorials (10h/year)*, M2 BIM, University Côte d'Azur, Nice - France.
- 2022-2023 **Statistics and Modeling**, *Tutorials (24h/year)*, 3rd year bachelor's degree, University Côte d'Azur, Nice - France.
- 2021-2022 **Analysis and Modeling**, *Tutorials (20h/year)*, 1st year bachelor's degree, University Côte d'Azur, Nice - France.
- 2013-2016 **Teaching mission proposed to PhD students (*Monitorat* - Tutorials 64h/year)**.
- Probability and Statistics 2**, *Tutorials*, 2nd year bachelor's degree in Mathematics and MIEF, University Paris 13, Villetaneuse - France.
- Probability and Statistics 1**, *Tutorials*, 2nd year bachelor's degree in Mathematics, University Paris 13, Villetaneuse - France.
- Probability and Statistics**, *Tutorials*, 1st year Engineering degree (Apprentissage énergétique), Engineering School Sup Galilée, Villetaneuse - France.
- Statistics**, *Tutorials*, 1st year Engineering degree MACS (Mathématiques Appliquées et Calcul Scientifique), Engineering School Sup Galilée, Villetaneuse - France.
- Probability**, *Tutorials*, 2nd year IUT-Info, University Paris 13, Villetaneuse - France.
- Inferential statistics**, *Tutorials*, 2nd year DUT-GEA, University Paris 13, Bobigny - France.

Review Activities

- PC Member for the ECAI 2024 (Outstanding PC Member Award), ECAI 2025.
- Computers in Biology and Medicine (Elsevier), Vaccine (Elsevier), Medical Image Analysis (Elsevier), Neuroimage (Elsevier), SMAI J. of Computational Mathematics

Softwares

- [MC²VAE](#) Variational causal disentanglement from multimodal data: design and code.
- [Fed-mv-PPCA](#) Bayesian framework for federated multi-view heterogeneous datasets assimilation based on Probabilistic PCA: design and code.
- [Fed-BioMed](#) Open-source federated learning framework: participation to research and development.

Responsibilities and Management

- Administration** In charge of the pedagogical organization of the AI for Health track of the [Data Science&AI](#) (DSAI) Master 2 since 2023, Université Côte d'Azur, Sophia Antipolis - France
- SC** Part of the scientific committee of the GIS (scientific interest group) [FC3R](#) since Jul. 2023.
Part of the scientific committee of the [Academy 2 \(Complex Systems\)](#) since Nov. 2023.
Part of the scientific committee of [IABM26](#).
- OC** Member of the organizing committee of the [Complex days](#), Nice, Feb. 2025.
Member of the organizing committee of the spring school [GeMSS/Statlearn](#), Sophia Antipolis, Apr. 2025.
- Juries** Member of the admissibility jury for the 2025 recruitment campaign for researchers, Inria Center at University Côte d'Azur.
Member of the PhD jury of Aymene Bouayed (ENS Ulm, Paris - May 2026).
Member of the PhD jury of Irene Garcia Fogeda (University of Antwerp, Antwerp, Belgium - September 2026).
Member of the PhD jury of H. Liu (Paris - May 2024).
Member of the PhD jury of V. Montalibet (Bordeaux - September 2024).
Member of the DSAI Master 2 jury (recruitment and evaluation) since 2023.
- CSI** Member of the *Comité de Suivi Individuel* for the following PhD students:

Evariste Njomgue Fotso, Multimodal learning for sudden cardiac death risk prediction. Started in 2023

Amel Bakhouch, Analysis of European National Health data to better understand the outcomes of patients with vascular diseases. Started in 2025

National & International Projects and Collaborations

2026-now [Graph-Cure](#), *Inria-UCL associate team*.

Role: Team co-leader

Academic: A. Calissano, K. Diaz-Ordaz, University College London (UCL) (UK).

Clinic: R. Martinez-Nunez, Guy's Hospital, King's College London.

2025-now **CausalGene**, *CHAI - New to causality*, 3 UK partners, 1 French.

Role: French PI

Clinic: Guy's Hospital (UK).

Charitable company: [Asthma+Lung UK](#) (UK).

2024-now **TALENT**, *RHU France 2030 programme*, 8 French partners, 1 USA.

Role: WP co-leader, Multimodal risk prediction model

Clinic: Bordeaux University Hospital (France).

2021–2025 **SimCardioTest**, *EU H2020 programme*, 10 organizations, 6 EU countries and USA.

Role: WP leader, In-silico trials & Data science

Academic: Center for Research and Innovation in Bioengineering (Ci2B), [Universitat Politècnica de València](#) (Spain).

Industry: [InSilicoTrials](#) (Italy).

2018–2019 **DYNAM-HIC**, *Inria Associate team*, France and USA.

Academic: [Harvard Program for evolutionary Dynamics](#) (USA).

2017–2019 **Ebovac**, *IMI Ebola+ programme*, 6 partners, EU, UK, Sierra Leone, Burkina Faso.

Academic: [Oxford Vaccine Group](#) (UK).

Industry: [Janssen Vaccines & Prevention B.V.](#) (Belgium).

Clinic: Vaccine Research Institute ([VRI](#), France).

2013–2016 **Inflamex**, *Laboratory of Excellence (LabEx)*, France.

Clinic: Inserm UMR-U978 Signaling Adapters in Hematology ([ASIH](#), France).

Outreach activities

2025 [AI'm in France](#), Why do researchers from around the world choose France for their research in #IA? Video for IA Summit 2025, Paris, France.

2022 [1 scientifique 1 classe: Chiche!](#), to encourage the pursuit of scientific studies and explain research activities to high school students. 3 classes, Lycée Tocqueville, Grasse, France.

Computer Skills

Programming **Python, Matlab, R**

Editing & Office **Office, L^AT_EX, OpenOffice**

Operating Systems **Mac OSX, Linux, Windows**

Others **Monolix, IdentifiabilityAnalysis (Mathematica), DAISY (Reduce3.8)**

Languages

Italian **Mother tongue**

English **Fluent**

French **Bilingual**

Spanish **Fluent**